

Todo Planner

A Modern To-Do and Personal Planning Web Application

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Agenda

1. Overview
2. Architecture
3. The Application
4. Engineering

Overview

What is Todo Planner?

A task-management and personal-planning web application.

It solves one well-scoped problem: giving a person a reliable place to record what they need to do, and to see what deserves attention *now*.

- Next.js 15 + React 18 + TypeScript
- PostgreSQL via Supabase
- 79 source files, 13,338 lines
- 12-table relational schema



Core Capabilities

- **Tasks** — priority, status, dates
- **Projects** — grouped workstreams
- **Tags** — colour-coded labels
- **Scheduling** — overdue / today / upcoming
- **Subtasks** — nested checklists
- **Dependencies** — blocking relationships
- **Recurrence** — repeating tasks
- **Reminders** — scheduled notifications
- **Search** — full-text filtering

Architecture

Technology Stack

Layer	Technology	Role
Framework	Next.js 15.0.7	App Router, server actions
UI	React 18.3.1	Component model
Language	TypeScript 5.3.3	Static typing
Styling	Tailwind CSS 3.4.1	Utility-first, dark mode
Database	PostgreSQL (Supabase)	Twelve tables
Server state	TanStack Query 5.17	Caching, optimistic updates
Client state	Zustand 4.4.7	Local UI state
Auth	jose + bcryptjs	Custom JWT sessions

Server actions, not REST

Data logic is concentrated in [eight server-action modules](#) under `src/lib/actions/` — `auth`, `tasks`, `subtasks`, `projects`, `tags`, `reminders`, `recurrence`, `dependencies`.

Client components call these directly. TanStack Query wraps them for caching and [optimistic updates](#) — a new task appears before the round trip finishes.

Only three API routes exist, for jobs server actions cannot serve:

<code>/api/heartbeat</code>	Keep-alive ping for Supabase
<code>/api/reminders/check</code>	Scheduled reminder processing
<code>/api/import-pfa-tasks</code>	One-off bulk import

Authentication

A purpose-built session layer, giving the application direct control over the user record:

- Passwords hashed with `bcrypt`, never stored in the clear
- Signed JWT issued via `jose`, held in an `HTTP-only cookie` — inaccessible to client-side scripts
- 24-hour session lifetime
- Every authenticated route guarded by `src/middleware.ts`

Why build it

Owning the session layer keeps the `users` table a first-class part of the schema, so accounts join naturally against projects, tasks and tags.

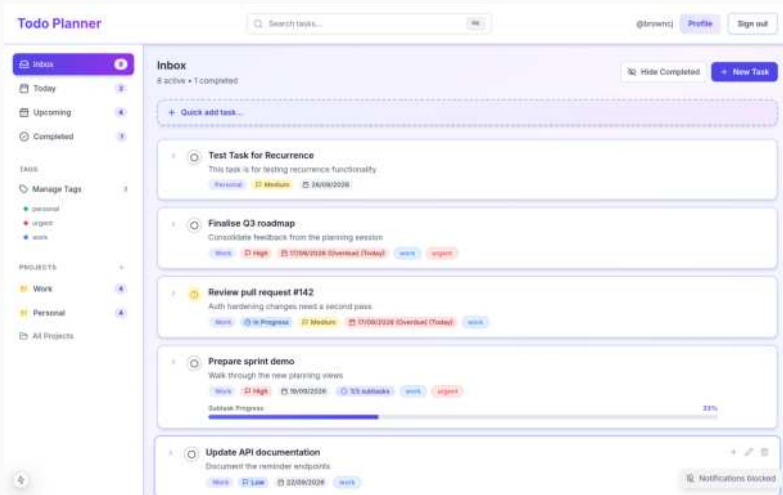
Data Model — 12 tables, 6 migrations

Identity	users, sessions, password_reset_tokens, profiles
Core	tasks, projects, subtasks
Labelling	tags, task_tags
Scheduling	recurrence, reminders
Graph	task_dependencies

tasks carries [ten indexes](#), including GIN indexes over to_tsvector of title and description for full-text search, plus partial indexes matching the exact queries each view issues.

The Application

The Inbox — every open task



Subtasks — nested checklists with progress

The screenshot displays the 'Todo Planner' application interface. At the top, there's a header with the app name, a search bar, and user profile links. The left sidebar contains navigation options like 'Inbox', 'Today', 'Upcoming', 'Completed', 'Tags', 'Manage Tags', 'Projects', and 'All Projects'. The main area shows a list of tasks in the 'Inbox'. Each task has a title, description, tags, and a due date. Some tasks have subtasks, which are shown as a list below the main task. The 'Prepare sprint demo' task includes a progress bar for its subtasks, showing 11% completion. The 'Update API documentation' task is at the bottom of the list.

Todo Planner Search tasks... @brywnc Profile Sign out

Inbox 8 active • 1 completed Hide Completed + New Task

+ Quick add task...

- Test Task for Recurrence**
This task is for testing recurrence functionality
Personal Medium 28/09/2028
- + Add first subtask
- Finalise Q3 roadmap**
Consolidate feedback from the planning session
Work High 01/10/2028 Overdue (Today) work urgent
- Review pull request #142**
Auth hardening changes need a second pass
Work In Progress Medium 07/09/2028 Overdue (Today) work
- Prepare sprint demo**
Walk through the new planning views
Work High 19/09/2028 3/3 subtasks work urgent
Subtask Progress 11%
- Update API documentation**
Document the rest API endpoints

Notifications blocked

Today — overdue, due today, unscheduled

The screenshot displays the 'Todo Planner' web application. The interface includes a sidebar on the left with navigation links for 'Inbox', 'Today' (selected), 'Upcoming', and 'Completed'. Below these are sections for 'TAGS' (personal, urgent, work) and 'PROJECTS' (Work, Personal, All Projects). The main content area is titled 'Today' and shows a summary of '3 tasks • 1 completed today'. It features a 'Quick add task...' input field. Tasks are listed in three sections: 'Overdue (1)' with 'Renew gym membership' (due 14/09/2026), 'Due Today (2)' with 'Finalise Q3 roadmap' (due 11/09/2026) and 'Review pull request #142' (due 17/09/2026), and 'No Due Date (1)' with 'Plan weekend trip'. Each task card includes a title, description, priority, category, and status tags. A 'Sign out' button is in the top right, and a 'Notifications blocked' indicator is in the bottom right.

Todo Planner

Search tasks...

@brywnj Profile Sign out

Inbox 3

Today 3

Upcoming 4

Completed 3

TAGS

Manage Tags 3

- personal
- urgent
- work

PROJECTS

- Work 4
- Personal 4
- All Projects

Today

3 tasks • 1 completed today

+ New Task

+ Quick add task...

Overdue (1)

- ☐ Renew gym membership
Annual plan expires this week
Personal Medium 14/09/2026 Overdue personal

Due Today (2)

- ☐ Finalise Q3 roadmap
Consolidate feedback from the planning session
Work High 11/09/2026 Overdue Today work urgent
- ☒ Review pull request #142
Auth hardening changes need a second pass
Work In Progress Medium 17/09/2026 Overdue Today work

No Due Date (1)

- ☐ Plan weekend trip
Shortlist two destinations
Personal Low personal

Notifications blocked

Upcoming — grouped by calendar day

The screenshot displays the 'Upcoming' view of a 'Todo Planner' application. The interface is organized into a sidebar on the left and a main content area on the right.

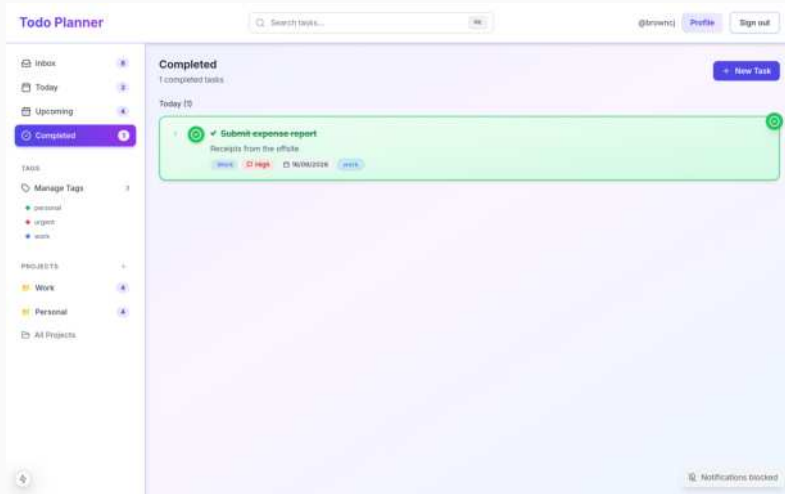
Sidebar:

- Navigation:** Includes 'Inbox' (3), 'Today' (3), 'Upcoming' (4, highlighted), and 'Completed' (3).
- TAGS:** Includes 'Manage Tags' (3), 'personal' (green), 'urgent' (red), and 'work' (blue).
- PROJECTS:** Includes 'Work' (4, yellow), 'Personal' (4, orange), and 'All Projects'.

Main Content Area:

- Header:** 'Upcoming' with '4 upcoming tasks' and a '+ New Task' button. A search bar 'Search tasks...' is at the top right.
- Quick Add:** A dashed box with '+ Quick add task...'.
- Task Grouping:** Tasks are grouped by day:
 - Saturday, 19/09/2026 (1):** 'Prepare sprint demo' (Walk through the new planning views). Subtasks: 'Work' (High), '1/3 subtasks', 'Work', 'Urgent'. Subtask Progress: 33%.
 - Monday, 21/09/2026 (1):** 'Book dentist appointment' (Personal, Low, 21/09/2026, Personal, Urgent).
 - Tuesday, 22/09/2026 (1):** 'Update API documentation' (Document the reminder endpoints). Subtasks: 'Work' (Low), '22/09/2026', 'Work'.
 - Saturday, 26/09/2026 (1):** 'Test Task for Recurrence' (This task is for testing recurrence functionality). Subtasks: 'Personal' (Medium), '26/09/2026'.
- Footer:** A 'Notifications blocked' icon in the bottom right corner.

Completed — a record of finished work



Projects — completion tracked per workstream

The screenshot displays the 'Todo Planner' interface. On the left sidebar, there are sections for 'Inbox', 'Today', 'Upcoming', 'Completed', 'TAGS' (personal, urgent, work), and 'PROJECTS' (Work, Personal, All Projects). The 'Work' project is selected. The main area shows a 'Back to Projects' link, a 'Work' project header with a progress bar (5 tasks, 20% complete), and a list of tasks: 'Finalise Q3 roadmap', 'Review pull request #142', 'Prepare sprint demo', and 'Update API documentation'. Each task has a description, tags, and a due date. A 'Notifications blocked' message is visible at the bottom right.

Todo Planner Search tasks... @brywncj Profile Sign out

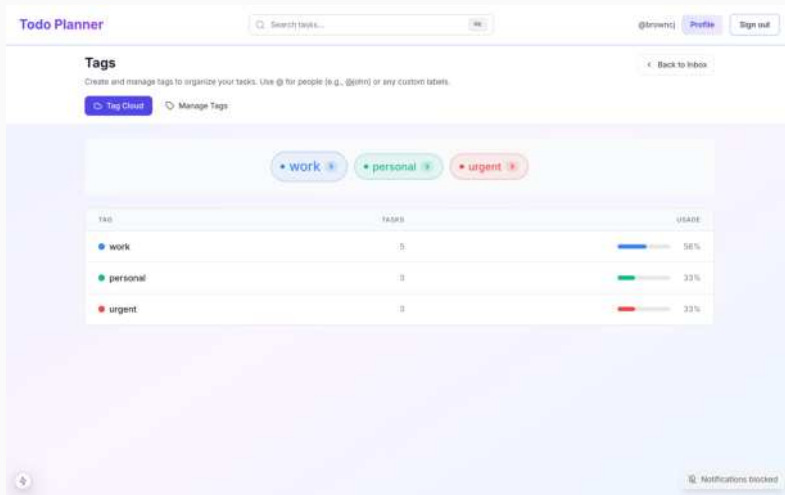
Work Team and delivery work 5 tasks • 20% complete

+ Quick add task...

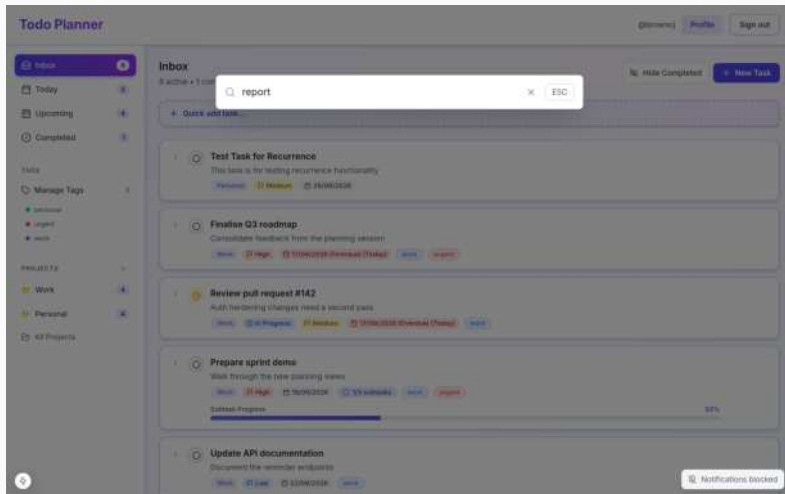
- Finalise Q3 roadmap**
Consolidate feedback from the planning session
Work High 11/08/2028 (Overdue) (Today) work urgent
- Review pull request #142**
Auth hardening changes need a second pass
Work In Progress Medium 07/08/2028 (Overdue) (Today) work
- Prepare sprint demo**
Walk through the new planning views
Work High 16/08/2028 1/3 subtasks work urgent
Subtask Progress 23%
- Update API documentation**
Document the reminder endpoints
Work Low 22/08/2028 work

Notifications blocked

Tags — cloud and usage statistics



Search — filters as you type



Engineering

Area	Files	Lines
src/components	26	4,589
src/lib	18	3,589
src/app	24	3,200
src/types	3	970
src/hooks	7	980
Total	79	13,338

Healthy distribution — component layer largest, logic concentrated in `src/lib`, no single file dominating.

Verified

- `npm run lint` passes
- `CI=1 npm run build` passes
- End-to-end run with **zero** server errors

By design

- TypeScript throughout
- Query-matched database indexes
- Data logic in one layer
- Optimistic UI updates

The application was launched and driven end to end for this presentation. Every screenshot comes from that running instance.

A coherent architecture

Server actions concentrate data logic in one place. The schema is properly indexed for the queries the interface actually issues. The interface is clean and consistent across every view.

A complete personal workflow

Capture, organise, schedule, break down, and review — Todo Planner covers the full loop, from a task typed into the inbox to a record of finished work.